Scientific notation: a number multiplied by a power of 10

Coefficient

A number **from 1 to 9.99…**
Only **ONE digit** (not zero) before the comma.

$$\underline{4{,}5} \cdot 10^{4}$$

Exponent

Positive (+) → big number (≥ 10)
Negative (−) → small number (< 1)



$1{,}5 \cdot 10^{8}$ km
distance Earth–Sun



$2 \cdot 10^{-6}$ m
size of a bacterium

✗

$45 \cdot 10^{3}$
wrong: two digits before the comma

✗

$0{,}45 \cdot 10^{5}$
wrong: the first digit is a zero